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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	VBall
Selected Domain	Domain 2: Orbyte Smart SME Challenge, "Less Busywork, Better Business"
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	SMErgers
Name (Leader)	Vincent Chew Wen Jie
Email & Phone (Leader)	vincent0512@student.usm.my | +6011-20208512
University Affiliation(s)	Universiti Sains Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small event companies in Malaysia run every job through WhatsApp. After the quote is signed, the client and vendors keep changing it in the group chat: 15 more guests, a bigger LED wall, an extra hour of overtime, two emergency microphones on the night. Nobody turns those messages into a record. When the job ends, someone scrolls through weeks of chat to rebuild the final bill, extras go unbilled, and when the client says "I never agreed to that", there is nothing to show, so the owner often gives the money away to keep the client.

Who is affected: owner-managed event companies of about 5 to 30 staff, their project managers and finance staff, and their clients, who also remember a different version. The problem comes from a team member's experience at an event company, where it happened on almost every event. It is one person's account, not a survey.

Why it matters: micro, small and medium enterprises are 96% of Malaysian businesses and contribute 39.7% of GDP (DOSM). For a small firm, unbilled extras come straight out of thin margins, and disputes cost time. The cause sits inside the firm's own process, so software can fix it. No public data exists for event companies, so the pilot measures the real rate.

In our simulated job, 10 changes worth RM3,860 sat unrecorded in 35 messages.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SMErgers reads a job's WhatsApp group as messages arrive and turns what was agreed into approved changes and a correct draft bill. It follows the Domain 2 journey: a message arrives (trigger). It is matched to the signed quote, rate card and company directory (connected data); a reader finds item, quantity, price and agreement (decision). The project manager checks it, the project director signs off on large or unusual items, and the client confirms from a link (human approval). The new quote version is posted back to the group, with a draft bill, CSV and activity log (outcome). Every bill line links to the exact messages it came from. A test connector reads a group directly; production uses the official API.

Where AI fits: the brief says AI only where it adds value. An open language model on RunPod (Llama 3.1 8B, no training by us) reads the messy chat and cites the messages; plain code checks every price, rules and people decide, and the rules reader is the fallback. On three chats written blind by independent agents, it fully got 17 of 29 changes right against 13 for the rules (21 after fixing its errors, not independent), a small synthetic sample. The learning part is already built: from a company's past jobs the system counts what usually changes late (for dinners, overtime in 4 of 4) and suggests what to quote upfront. Every approval and dispute is a human-checked label, for training after a pilot, never on made-up data.

Workshop skills: cloud deployment on Replit; structured data models with a tamper-evident activity log; the WhatsApp Business Cloud API webhook format. No hosted GenAI API is called.

Nothing moves money or files anything: the bill is a draft for accounting software, and all prototype data is synthetic.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Stack: Python 3.12, Flask and Jinja templates; in-memory data store in the prototype (SQLite or PostgreSQL for a pilot); WhatsApp Business Cloud API webhook format, fed with synthetic messages, plus an optional local Node.js test connector that reads a test group directly; Replit for development and hosting; GitHub (Apache 2.0); pytest, 290+ automated tests. No hosted GenAI API is used; the AI reader is an open model on our own RunPod endpoint.

Block diagram (text):

WhatsApp group (webhook, test connector) or chat export → reader (rules; AI reader on RunPod when chosen) → approval workflow (PM → director → client) → quote versions and messages back to the group → draft bill, CSV, API, activity log. Inputs to the reader: signed quote, rate card, company directory, past jobs.

Official SDC sponsor-provided GenAI API or cloud platform: Replit (provided to participants by SDC) is used for development and hosting. No GenAI API is used. RunPod (sponsor credit) runs the AI reader's model on a GPU through a Serverless endpoint.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: one teammate worked at an event company and described the problem first-hand; mentor sessions added who uses each output. Define: after rejecting a payment-fraud guard (rare, hard to sell), a notifications app (no finished result) and chasing late payments (cause outside the firm), we locked one problem: agreements in chat never become a record. Ideate: we mapped each cause to one feature and cut anything that did not produce a finished result. Prototype: a working journey from message to quote version. Test: a first-time-user review showed too many clicks and technical labels, so we added a guided five-step hand-over, one-click bulk checks and plain words; role workspaces and a new-project form followed. 290+ automated tests cover parsing, rules, permissions, the bill and the pages.

Deployment: the app runs on Replit with synthetic data and a simulated WhatsApp webhook. For Phase 2 we will re-test the reader on chats written by our teammates, time the final bill by hand against SMErgers, and prepare the official WhatsApp API connection. We have not yet tested with real event companies.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**8.3** "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises." Also **8.2** "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In our simulated dinner job, 10 changes worth RM3,860 (11.2% of the RM34,530 quote) were agreed in 35 chat messages, with 0 recorded. Working estimate for event companies today: about 75% of changes agreed in chat reach the final bill (team member's experience, unverified). The pilot measures the real figure. No public figure exists for event companies. MSMEs are 96.2% of Malaysian establishments and 39.7% of GDP (DOSM, MSME Performance 2025).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Share of changes agreed in chat that appear on the final bill (billed ÷ agreed, %), and minutes to build the final bill.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 95% of agreed changes on the final bill (from the 75% estimate), and a 50% cut in time to build the final bill, measured on a timed manual test against SMErgers.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask 3.1.3	code	BSD-3-Clause
Jinja2, MarkupSafe, Werkzeug (Flask dependencies)	code	BSD-3-Clause
tzdata 2026.5	code (time-zone data)	Apache-2.0
pytest 9.1.1	code (tests)	MIT
Llama 3.1 8B Instruct (Meta), used as it is, no training by us	pre-trained model	Llama 3.1 Community License
vLLM worker on RunPod Serverless (serves the model)	code	Apache-2.0
Ollama (local development only)	code	MIT
whatsapp-web.js ^1.34.7 and its Puppeteer dependency (optional local test connector, unofficial, not part of the product)	code	Apache-2.0

One pre-trained model is used as it is, with no training by us: Llama 3.1 8B (Meta) on our RunPod endpoint, as the AI reader (inference only). No external dataset is used. All data in the application is synthetic and written by the team; the blind test chats were written by separate AI agents.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

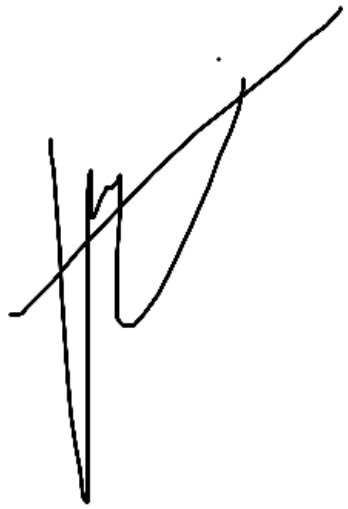
6.2 (continued): further third-party items

qrcode (npm) ^1.5.4, test connector pairing page	code	MIT
WhatsApp Business Cloud API (Meta), design and webhook format only	API specification	Meta Platform Terms
Instrument Serif, Hanken Grotesk, DM Mono (pitch deck only)	fonts	SIL Open Font License
Unsplash photos (pitch deck only)	images	Unsplash License
DOSM MSME statistics (cited, not used as a dataset)	public statistics	DOSM terms of use

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24th October 2026